Big Data Processing and Analytics

Teachers

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Office hours

- Class-time (break, end of lesson)
- Or send an e-mail for an appointment

Weekly schedule

- Lectures (45 hours)
  - Friday - 16:00-19:00 – Room 6
- Practices (15 hours)
  - Thursday – 13:00-14:30 – LAIB1 - Team 1 (A-K)
  - Friday - 14:30-16:00 – LAIB1 - Team 2 (L-Z)
  - There are **no lab activities** during the **first two weeks**

Practices

- We will provide you a specific account on the BigData@Polito cluster
  - https://jupyter.polito.it
- Detailed information will be provided next week
  - You will receive an email from the administrator of the cluster with username and password

Topics

- Lectures
  - Introduction to Big data
  - Hadoop
    - Architecture
    - **MapReduce programming paradigm**
  - Spark
    - Architecture
    - **Spark programs based on RDDs (Resilient Distributed Data sets) and Spark SQL (DataFrames)**

Topics

- Data mining and Machine learning libraries for Big Data
  - **MLlib** (Apache Spark's scalable machine learning library)
- Streaming data analysis
  - **Spark Streaming**

Topics

- Laboratory activities
  - Application development on Hadoop and Spark

Prerequisites / prior knowledge

- Basic programming skills
  - **Java language (MapReduce for Hadoop)**
  - **Python language (Spark)**
- Basic knowledge of traditional database concepts (recommended)
  - Relational data model
  - SQL language

Material

- Web page
  - https://minds.polito.it/courses/01dshyg/
  - Slides, exercises, lab activities, ..
- Video lectures/Virtual classrooms
  - On the Teaching portal
    - https://didattica.polito.it

Books and Readings

- Reference books:
  - Matei Zaharia, Bill Chambers. Spark: The Definitive Guide (Big Data Processing Made Simple). O'Reilly Media, 2018.
  - Advanced Analytics and Real-Time Data Processing in Apache Spark. Packt Publishing, 2018.
  - Matei Zaharia, Holden Karau, Andy Konwinski, Patrick Wendell. Learning Spark (Lightning-Fast Big Data Analytics). O'Reilly, 2015.
  - Tom White. Hadoop, The Definitive Guide. (Third edition). O'Reilly Media, 2015.
  - Donald Miner, Adam Shook . "MapReduce Design Patterns: Building Effective Algorithms and Analytics for Hadoop and Other Systems." O'Reilly, 2012

Exam rules

- Written exam
  - 2 programming exercises (max 27 points)
    - Design and develop programs based on the Hadoop MapReduce programming paradigm and/or Spark APIs
  - 2 questions/theoretical exercises (max 4 points)
    - Topics
      - Technological characteristics and architecture of Hadoop and Spark
      - HDFS
      - MapReduce programming paradigm
      - Spark RDDs, transformations and actions
      - Spark SQL
      - Spark Streaming
      - Spark MLlib

Exam rules

- On-site written exam on the **Exam platform with Lockdown browser** – **You must bring your own PC**
  - 1.5 hours
  - The exam is **open book**
    - Books, notes, and any other paper material are allowed
    - Electronic devices of any kind (PC, mobile phone, calculators, etc.) are not allowed, except the PC used for the exam itself
- Exam examples are available on the web page of the course